

Fig. 3: Selection of 802.11b preamble

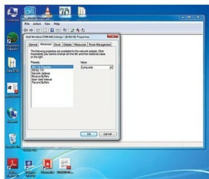


Fig. 4: Adhoc mode

bles use of available radio to select other nodes by long and short or long-only options. In the wireless network, adhoc mode can also be selected by enabling it. Even though many settings are available, the above-discussed are important.

Windows 7 enables easy-to-setup network using LAN or wireless connection. Fig. 5 shows Network Connection control, where the option for a type of network can be chosen such as home, private or public.

Fig. 6 shows the network of Reliance in use and in the box all other networks available in the area.

Windows 7 is better than Windows XP as it provides more options for connecting system devices such as a modem. In Windows 10, speed monitor is available but it is equivalent to Windows 7 add-on app or software like Acrylic.

Wi-Fi Manager app

The available channels and their signal strength surrounding host user can be managed. If operating frequen-

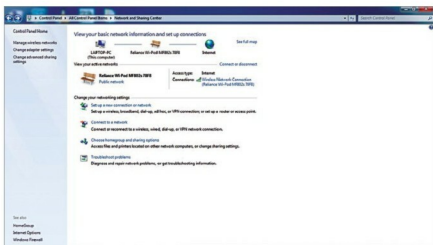


Fig. 5: Windows 7 network connections



Fig. 6: Windows 7 available network connections

cies are multiple, it will show which operating link (2.4GHz or 5GHz) is being used for communication.

Channel number in which connection sustains is seen and can be programmed. In case of non-overlapping channel, signal strength would be -108dbm. The other channel's strength would be near -29dbm only.

Fig. 7 shows the channel number and overall network parameters. The area and time of connection determine link quality.

Simulation results

Operating system services such as registry and cmd mode instruction such as ping will show data speed from a particular website. Configurations such as header and data

packet size are kept default.

Showing network connectivity between devices will tell the strength of the link. The simulator used will tell the following:

1. List of neighbours
2. List of users connected to the hot spot

3. Assigned channel number
The experimental setup shown in Fig. 10 has a monitor, Macbook Pro, keyboard and mouse in which TP-link sys modem is configured to connect to the gateway through the nearest AP. Unlike Wi-Fi Sense, our simulator will show the following:

Access points. More Wi-Fi network information (SSID/BSSID) and connected users around us

Signal level. High signal quality